

Question_9:

Code:

```
1  /*
2  Question No: 9
3  Given a rooted tree where every node contains a lowercase
4  letter, answer queries u c.
5  For each query, count the nodes in the subtree of u
6  whose character is c.
7  */
8  #include <iostream>
9  #include <vector>
10 #include <algorithm>
11 using namespace std;
12
13 vector<int> g[100005];
14 vector<int> pos[26];
15
16 int tin[100005], tout[100005], timer=0;
17 string s;
18 void dfs(int u, int p){
19     tin[u] = ++timer;
20     pos[s[u-1] - 'a'].push_back(timer);
21
22     for(int v: g[u])
23         if(v != p)
24             dfs(v, u);
25     tout[u] = timer;
26 }
27 int main(){
28     int n, q;
29     cin >> n >> q;
30     cin >> s;
31     for(int i=0; i<n-1; i++){
32         int u, v;
33         cin >> u >> v;
34         g[u].push_back(v);
35         g[v].push_back(u);
36     }
37     dfs(1, 0);
38     while(q--){
39         int u;
40         char c;
41         cin >> u >> c;
42         auto &v = pos[c - 'a'];
43
44         cout << upper_bound(v.begin(), v.end(), tout[u])
45              << -lower_bound(v.begin(), v.end(), tin[u]) << "\n";
46     }
47 }
```

Output:

Test Case_1:

```
3 1
aba
1 2
1 3
1 a
2
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
4 1
WSX
1 2
1 3
1 1
1 w
1
PS C:\Users\Prabin Yadav\
```